

# Shi Jie Samuel Tan

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## INTRODUCTION

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I am a PhD student at the Joint Center for Quantum Information and Computer Science (QuICS) at the University of Maryland, College Park. I am broadly interested in quantum error correction (QEC) and fault tolerance. I utilize concepts and tools in quantum information theory, homological algebra, and graph theory to understand and advance the construction of quantum error correcting codes and fault-tolerant protocols. I also dabble in quantum algorithms and quantum metrology.

## EDUCATION

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### University of Maryland, College Park

Ph.D. in Computer Science; Advisors: Daniel Gottesman and Michael J. Gullans

College Park, MD

2023–2027 (Expected)

### Haverford College

B.S. in Computer Science and Physics

Magna Cum Laude; High Honors in Computer Science and Physics

Haverford, PA

2019–2023

## WORK

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### University of Maryland, College Park

Research Assistant; Supervisors: Daniel Gottesman and Michael J. Gullans

College Park, MD

2023–Current

### Google Quantum AI

Student Researcher; Mentors: Noah Shutty and Michael Newman

Playa Vista, CA

Fall 2026

### IBM Quantum

QEC Theory Research Intern; Mentors: Guanyu Zhu and Theodore J. Yoder

Yorktown Heights, NY

Summer 2026

### University of Maryland, College Park

Grader and Teaching Assistant

College Park, MD

2025–2026

### QuEra Computing Inc.

QEC Research Intern; Mentors: Hengyun (Harry) Zhou and Arpit Dua

Boston, MA

Summer & Fall 2025

### Los Alamos National Laboratory

Quantum Algorithms Research Fellow; Mentors: Yiğit Subaşı and Andrew Sornborger

Los Alamos, NM

Summer 2023

### California Institute of Technology

QEC SURF Fellow; Mentors: John Preskill and Christopher A. Pattison

Pasadena, CA

Summer 2022

### University of Maryland, College Park

REU-CAAR Summer Researcher; Mentors: Matthew Coudron and Gorjan Alagic

College Park, MD

Summer 2021

PUBLICATIONS & PREPRINTS

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♦ denotes equal contribution. ♠ denotes alphabetical order.

- [1] **S.J.S.T.**♦, Ian Gill♦, Eric Huang♦, Pengyu Liu, Chen Zhao, Hossein Dehghani, Aleksander Kubica, Hengyun Zhou, and Arpit Dua, “Generalized matching decoders for 2D topological translationally-invariant codes”, [arXiv preprint arXiv:2603.05402 \(2026\)](#).
- [2] Pengyu Liu, **S.J.S.T.**, Eric Huang, Umut A Acar, Hengyun Zhou, and Chen Zhao, “Achieving Optimal-Distance Atom-Loss Correction via Pauli Envelope”, [arXiv preprint arXiv:2603.04156 \(2026\)](#).
- [3] **S.J.S.T.**♦, Yifan Hong♦, Ting-Chun Lin♦, Michael J Gullans, and Min-Hsiu Hsieh, “Single-Shot Universality in Quantum LDPC Codes via Code-Switching”, [arXiv preprint arXiv:2510.08552 \(2025\)](#), [Accepted as a plenary talk at QIP 2026].
- [4] ♠ Noah Berthussen, Michael J. Gullans, Yifan Hong, Maryam Mudassar, and **S.J.S.T.**, “Automorphism gadgets in homological product codes”, [arXiv preprint arXiv:2508.04794 \(2025\)](#).
- [5] Noah Berthussen, **S.J.S.T.**, Eric Huang, and Daniel Gottesman, “Adaptive Syndrome Extraction”, [PRX Quantum 6, 030307 \(2025\)](#), [Accepted as a talk at TQC 2025].
- [6] **S.J.S.T.** and Lev Stambler, “Effective Distance of Higher Dimensional HGP’s and Weight-Reduced Quantum LDPC Codes”, [Quantum 9, 1897 \(2025\)](#).
- [7] Dhruvil Patel♦, **S.J.S.T.**♦, Yiğit Subaşı, and Andrew T. Sornborger, “Optimal Coherent Quantum Phase Estimation Via Tapering”, [PRX Quantum 7, 020302 \(2026\)](#).
- [8] **S.J.S.T.**, Christopher A. Pattison, Matt McEwen, and John Preskill, “Resilience of the surface code to error bursts”, [Phys. Rev. A 113, 022450 \(2026\)](#).
- [9] **S.J.S.T.**, Huyen Trang Dang, Sarah Keim, Maja Bucan, and Sara Mathieson, “Identity-by-descent (IBD) segment outlier detection in endogamous populations using pedigree cohorts”, [bioRxiv, 2024–08 \(2024\)](#).
- [10] ♠ Yongtao Deng and **S.J.S.T.**, “Random Walks on the Generalized Symmetric Group: Cutoff for the One-sided Transposition Shuffle”, [arXiv preprint arXiv:2211.10462 \(2022\)](#).
- [11] Suchetan Dontha♦, **S.J.S.T.**♦, Stephen Smith, Sangheon Choi, and Matthew Coudron, “Approximating Output Probabilities of Shallow Quantum Circuits Which Are Geometrically-Local in Any Fixed Dimension”, in [Proceedings of the 17th Conference on the Theory of Quantum Computation, Communication and Cryptography \(TQC 2022\) \(2022\)](#), [Accepted as a talk at TQC 2022].
- [12] Kelly Finke, Michael Kourakos, Gabriela Brown, Huyen Trang Dang, **S.J.S.T.**, Yuval B. Simons, Shweta Ramdas, Alejandro A. Schäffer, Rachel L. Kember, Maja Bučan, and Sara Mathieson, “Ancestral haplotype reconstruction in endogamous populations using identity-by-descent”, [PLOS Computational Biology, edited by Degui Zhi, 10.1371/journal.pcbi.1008638 \(2021\)](#).

TALKS

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Invited talks are indicated by ★.

★ National University of Singapore (NUS) Centre for Quantum Technologies (CQT)

September 17, 2026

- ★ IBM QITEC Summer Workshop June 3, 2026  
*Single-Shot Universality in Quantum LDPC Codes via Code-Switching*
- ★ USYD-PsiQuantum Seminar April 16, 2026  
*Single-Shot Universality in Quantum LDPC Codes via Code-Switching*
- APS Global Physics Summit March 17, 2026  
*Single-Shot Universality in Quantum LDPC Codes via Code-Switching*
- ★ Harvard University (Lukin Group) March 6, 2026  
*Single-Shot Universality in Quantum LDPC Codes via Code-Switching*
- 29th Annual Quantum Information Processing Conference (**Short Plenary**) January 26, 2026  
*Single-Shot Universality in Quantum LDPC Codes via Code-Switching*
- ★ Freie Universität Berlin (Eisert Group) November 18, 2025  
*Single-Shot Universality in Quantum LDPC Codes via Code-Switching*
- Joint Quantum Institute (JQI) Friday Quantum Seminar (College Park, MD) October 3, 2025  
*Adaptive Syndrome Extraction*
- 20th Conference on the Theory of Quantum Computation, Communication and Cryptography September 18, 2025  
*Adaptive Syndrome Extraction*
- ★ National University of Singapore (NUS) Centre for Quantum Technologies (CQT) May 22, 2025  
*Automorphism gadgets in homological product codes*
- ★ University of Maryland Communication, Control and Signal Processing (CCSP) Seminar October 17, 2024  
*Effective Distance of Higher-Dimensional HGP's and Weight-Reduced Quantum LDPC Codes*
- Caltech SFP Summer Seminar Day August 18, 2022  
*Proving the existence of an accuracy threshold for the Union-Find decoder*
- University of Maryland Research Experience for Undergraduates Combinatorics and Algorithms August 13, 2021  
for Real Problems (REU-CAAR) Research Presentation  
*Simulation of Low-Depth Quantum Circuits*

## POSTERS

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- 8th International Conference on Quantum Error Correction June 11, 2026  
*Single-Shot Universality in Quantum LDPC Codes via Code-Switching*
- 8th International Conference on Quantum Error Correction June 11, 2026  
*Generalized matching decoders for 2D topological translationally-invariant codes*
- 8th International Conference on Quantum Error Correction June 9, 2026  
*Achieving Optimal-Distance Atom-Loss Correction via Pauli Envelope*
- Monte Verità Quantum Codes April 21, 2026  
*Single-Shot Universality in Quantum LDPC Codes via Code-Switching*
- 29th Annual Conference on Quantum Information Processing January 26, 2026  
*Automorphism gadgets in homological product codes*
- 7th International Conference on Quantum Error Correction August 12, 2025  
*Automorphism gadgets in homological product codes*
- 7th International Conference on Quantum Error Correction August 11, 2025  
*Adaptive Syndrome Extraction*
- 28th Annual Conference on Quantum Information Processing February 26, 2025  
*Effective Distance of Higher-Dimensional HGP's and Weight-Reduced Quantum LDPC Codes*
- 19th Conference on the Theory of Quantum Computation, Communication and Cryptography September 11, 2024

### *Tapered Quantum Phase Estimation*

- 6th International Conference on Quantum Error Correction November 1, 2023  
*Resilience of the surface code to error bursts*
- 13th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics August 7, 2022  
*Comparison of cohort-based identical-by-descent (IBD) segment finding methods for endogamous populations*
- 25th Annual Conference on Quantum Information Processing March 10, 2022  
*Approximating Output Probabilities of Shallow Quantum Circuits which are Geometrically-local in any Fixed Dimension*
- 7th–8th International Conference on Algorithms for Computational Biology November 9, 2021  
*Improvements to ancestral haplotype reconstruction in pedigrees*
- Research in Computational Molecular Biology – 25th Annual International Conference August 29, 2021  
*Improvements to ancestral haplotype reconstruction in pedigrees*
- Haverford College Marian E. Koshland Integrated Natural Sciences Center (KINSC) Undergraduate Science Research Symposium September 10, 2020  
*Validating Ancestral Haplotype Reconstruction In Endogamous Populations using Identical-by-Descent*

## FELLOWSHIPS AND AWARDS

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- National University of Singapore Development Grant 2024–2026
- MathQuantum Graduate Fellowship (University of Maryland) 2023
- QuICS Lanczos Graduate Fellowship (University of Maryland) 2023–2025
- Louis B. Green Prize in Physics and Astronomy (Haverford College) 2023
- Phi Beta Kappa 2023
- Los Alamos National Laboratory Quantum Computing Summer School Fellowship 2023
- Singloh Hsu Scholarship (Haverford College) 2023
- Computing Research Association (CRA) Outstanding Undergraduate Researcher Award Finalist 2023
- Caltech Summer Undergraduate Research Fellowship 2022
- Marian E. Koshland Integrated Natural Sciences Center Summer Scholarship (Haverford College) 2022
- An Zhu-Google University of Maryland REU-CAAR Fellowship 2021
- Brian Kovaric Fellowship (Haverford College) 2020
- Prime Minister’s Book Prize (Ministry of Education, Singapore) 2016

## OTHER ATTENDANCE

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- Los Alamos Quantum Computing Summer School June–August 2023

## TEACHING (AS TA OR GRADER)

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- Computer Science 858G (University of Maryland, College Park) Spring 2026  
*Quantum Error Correction and Fault Tolerance*
- Computer Science 657 (University of Maryland, College Park) Fall 2025  
*Introduction to Quantum Information Processing*

- Physics H302 (Haverford College) Spring 2023  
*Advanced Quantum Mechanics*
- Computer Science B340 (Bryn Mawr College) Fall 2022  
*Analysis of Algorithms*
- Physics H214 (Haverford College) Spring 2022  
*Introductory Quantum Mechanics*
- Computer Science H345 (Haverford College) Spring 2022  
*Theory of Computation*
- Mathematics Question Center Fall 2021–Spring 2022  
*Calculus, Linear Algebra, and Abstract Algebra*
- Physics H213 (Haverford College) Fall 2021  
*Waves and Optics*
- Mathematics H121 (Haverford College) Spring 2021  
*Multivariable Calculus*
- Computer Science H311 (Haverford College) Fall 2020  
*Computer Security: Attacks and Defenses*

## REVIEWING

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I have been invited to review articles and abstracts for the following conferences and journals.

- Annual Quantum Information Processing Conference (QIP)
- Conference on Theory of Quantum Computation, Communication and Cryptography (TQC)
- Physical Review X Quantum

## COMPUTER SKILLS

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Ordered from most to least familiar.

- **Languages:** Python, Julia, SageMath
- **QC toolkits:** Stim

## REFERENCES

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- **Dr. Daniel Gottesman**, Brin Family Endowed Professor in Theoretical Computer Science  
Department of Computer Science, the Institute for Advanced Computer Studies, and the Joint Center for Quantum Information and Computer Science, University of Maryland, College Park  
dgottesm@umd.edu
- **Dr. Michael J. Gullans**, Adjunct Assistant Professor  
Department of Physics, Department of Computer Science, and the Institute for Advanced Computer Studies,  
University of Maryland, College Park  
mgullans@umd.edu